

N,OH-Immobilized Cu₃ Clusters *in-situ* Reconstructed from Single-Atom Metals for CO₂ Electromethanation in Bicontinuous Mesochannels

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Abstract

Cu-based catalysts hold promise for electrifying CO₂ to produce methane, an extensively used fuel. However, the selectivity remains insufficient due to the lack of effective catalyst design strategies in governing complex reaction pathways. Herein, we introduce a design principle for carbon-supported Cu catalysts that combines engineering both the atomic-scale electronic properties of Cu sites and mesoscale local reaction microenvironment of carbon support to steer the selective CO₂ reduction pathway to methane. Block copolymer-derived carbon fiber support with interconnected mesopores is developed to offer H₂O-deficient and long-range confined mesochannels with prolonged transport path for CO intermediate, thereby suppressing hydrogen evolution and boosting deep CO₂ reduction. Importantly, *in-situ* X-ray absorption and Raman spectroscopy reveal the dynamic reconstruction of unique N₂OH-immobilized Cu₃ clusters from atomically dispersed Cu under realistic operating conditions. The site possesses moderate *CO adsorption affinity and a low barrier for *CO hydrogenation yet a high barrier for C–C coupling, enabling intrinsically selective CO₂-to-methane reduction. The *in-situ* formed N₂OH-Cu₃ clusters rooted in bicontinuous mesochannels achieve efficient methane generation with an unprecedented Faradaic efficiency of 74.2% at an industry-level current density of 300 mA cm⁻². This work sheds light on effective design principles for steering reaction pathways in complex interfacial catalytic systems.

Introduction

Electrifying CO₂ transformation using renewable energy offers a carbon-neutral solution to CO₂ recycling and fuel generation.¹⁻² Among various feedstocks from CO₂ reduction reaction (CO₂RR), two electron-enabled CO and HCOOH have approached the performance thresholds for industry implementation.³ Compared to lightly reduced products, deeply reduced products, such as hydrocarbons and oxygenates, are economically more attractive and hold significant market sizes.¹ Particularly, methane (CH₄), an extensively used fuel, is compatible with the existing storage and transportation infrastructure. Nevertheless, the current benchmarking performance of CO₂-to-CH₄ remains far below the applicable threshold owing to grand challenges associated with the complicated eight proton-coupled electron transfer (PCET) steps, adsorption/transformation of multiple carbonaceous intermediates, and competition with other reaction pathways beyond CH₄.⁴⁻⁶ The success in achieving viable CO₂ electromethanation relies on the development of high-performance electrocatalysts.

Earth-abundant copper (Cu) is considered the most promising catalyst for reducing CO₂ to hydrocarbons; however, the CH₄ formation competes with C₂ products.⁷⁻⁸ Modifying electronic properties of Cu by alloying and interfacial interaction can stabilize *CHO intermediate and selectively produce CH₄ with suppressed C–C coupling on large-size Cu nanoparticles,⁹⁻¹⁰ while the performance needs further improvement to meet the application need. Notably, single-atom Cu catalysts have shown great promises to steer reaction pathways in complex catalytic processes thanks to their great designability of electronic properties.^{5, 11-14} To stabilize atomically dispersed metal sites, nitrogen-doped carbons are desirable supports due to their remarkably tunable morphology, pore architecture, and electronic structure of active M–N_x moieties.¹⁵⁻²² Several pioneering works have suggested that carbon-supported isolated Cu atoms are not electrochemically stable under negative potentials of realistic CO₂RR conditions and would likely be reduced to metallic Cu species, acting as the catalytic site for CO₂ reduction to CO and C₂H₅OH.²³⁻²⁵ However, to the best of our knowledge, such dynamic reconstruction has not been demonstrated in the CO₂-to-CH₄ transformation yet, and

the atomic-level structure of potential-driven Cu sites and their catalytic mechanisms toward CH₄ formation remain ambiguous.

Besides the atomic structure of carbon-supported Cu sites, the local reaction microenvironment also plays a critical role in CO₂ reduction, which is highly associated with the pore architecture of carbon supports.^{17, 26-28} To drive CO₂-to-CH₄ conversion, optimization of CO₂/H₂O concentration and elongation of the transport path of CO intermediate are vital for suppressing the side hydrogen evolution reaction (HER) at large overpotentials where CH₄ formation occurs and promoting deep CO reduction, respectively.^{27, 29} Challengingly, such expectations cannot be readily realized on conventional carbon supports with deficient and short-range pores due to the significant fraction of H₂O warping within the catalyst surface and the rapid loss of CO intermediates from the surface. We hypothesized that constructing bicontinuous mesoporous carbon to host Cu sites could create H₂O-deficient yet CO₂-rich surfaces to restrain the HER, because H₂O penetration in the pore region of catalysts loaded on the hydrophobic gas diffusion layer (GDL) can be suppressed. Moreover, the interconnected mesochannels would offer an enhanced long-range confinement space for increasing the residence time of the CO intermediate towards enabling effective interactive contact with Cu sites,³⁰⁻³¹ thereby creating a deep reduction-favorable local microenvironment.

Taken together, we here sought to develop bicontinuous mesoporous carbon for supporting atomically dispersed Cu, aiming to engineer both active site structure and local reaction microenvironment toward efficient CO₂-to-CH₄ conversion. To this end, we implemented the molecule-level microphase separation of a polyacrylonitrile-block-poly(methyl methacrylate) (PAN-*b*-PMMA) block copolymer to create interconnected mesoporous carbon fiber (IPCF) supports with abundant, uniform, and bicontinuous mesochannels for stabilizing isolated Cu-N₃ moieties (Cu-N/IPCF). More importantly, *in-situ* X-ray absorption and Raman spectroscopy studies revealed the dynamic reconstruction of atomically dispersed Cu to nitrogen and hydroxyl-immobilized Cu₃ (N,OH-Cu₃) clusters during the operating CO₂ reduction conditions, being the actual active sites to promote electrochemical CO₂-to-CH₄ reduction, as elucidated by extensive experimental analyses

and theoretical calculations. Accordingly, the synergistic coupling of *in-situ* formed N₂O₂-Cu₂ sites and bicontinuous confined mesoreactors effectively suppresses hydrogen evolution and steers selective CO₂ electromethanation, endowing the newly designed catalyst with a CH₄ Faradaic efficiency of 74.2% at a commercially viable current density of 300 mA cm⁻².

Results and Discussion

Design and synthesis of catalysts

In CO₂ electrolysis at the gas-solid-liquid triple phase boundary of the gas diffusion electrode (GDE) (Figure 1a), H₂O is required as a proton source for hydrocarbons. For carbon-supported Cu catalysts, H₂O is rich on the catalyst surface in the case of conventional pore-deficient carbon (Figure 1b) and short-range macro/mesoporous carbon supports (Figure 1c), while the CO₂ concentration is limited due to its low solubility and reaction with electrolyte, particularly in alkaline conditions.³²⁻³³ Such CO₂-depleted and H₂O-rich interface results in overwhelming HER at large overpotentials that are needed to drive deep CO₂ reduction to CH₄.²⁹ To restrain the HER, we intended to design an IPCF support with suppressed H₂O penetration in the long-range mesopore region on the hydrophobic GDE, thus creating a large portion of H₂O-deficient and CO₂-rich active interface (Figure 1d). It is noting that a small amount of H₂O can still enter mesochannels along with CO₂ stream, providing proton source yet with suppressed HER competition. The long-range mesochannels could also promote CH₄ formation by confining CO intermediates in mesochannel reactors to boost its successive reduction over the Cu sites;³⁰⁻³¹ such benefits are not inherently owned by conventional carbon supports with sparse and short-range macro/mesopores.

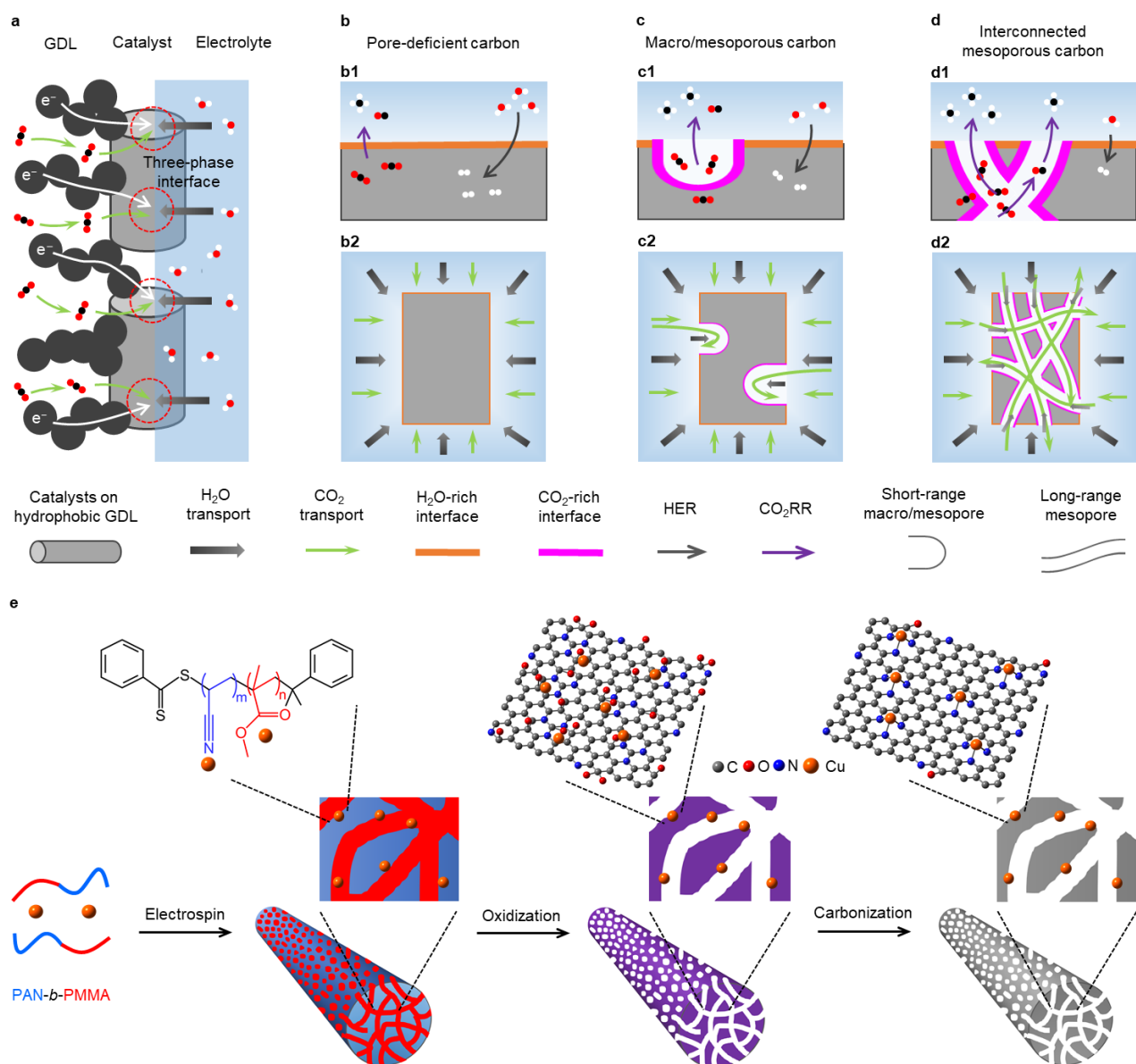


Figure 1. Design conception and catalyst synthesis. (a) Schematic showing GDE structure. (b-d) $\text{CO}_2\text{RR/HER}$ (b1, c1, d1) and $\text{CO}_2/\text{H}_2\text{O}$ transport (b2, c2, d2) properties at the three-phase interface in the local reaction microenvironment of mesopore-deficient carbon (b), macro/mesoporous carbon (c), and interconnected mesoporous carbon supports. (e) Synthesis procedures for the Cu-N/IPCF derived from the PAN-*b*-PMMA block copolymer.

To experimentally synthesize the as-proposed Cu-N/IPCF, a block copolymer-assisted approach was developed via the electrospinning, oxidation, and pyrolysis of the PAN-*b*-PMMA polymer containing cupric acetate (Figure 1e, Figure S1, Figure S2). Due to the microphase separation of the PAN-*b*-PMMA, this approach allows accurate control over the evolution of continuous PMMA domain networks rooted in the PAN matrix at the molecular level,³⁴ thus navigating the generation of the IPCF support upon the removal of the PMMA template and the formation of atomically

dispersed Cu–N₃ coordination by high-temperature thermal activation. The high homogeneity of single Cu sites enables us to probe their dynamic structure during CO₂RR. We have also synthesized another two types of catalysts as controls to justify the vital significance of bicontinuous mesopores. One is Cu–N₃ supported on mesopore-deficient carbon fiber (Cu-N/CF) using PAN alone (Figure S3). The other is Cu–N₃ supported on conventional porous carbon fiber with discontinuous macro/mesopores (Cu-N/PCF) using the physical PAN+PMMA blend.

Morphology and pore structure of catalysts

Owing to the absence of PMMA as pore templates, oxidizing Cu-PAN fibers and then transforming into the Cu-N/CF would not yield mesopores. This can be confirmed by scanning electron microscopy (SEM), transmission electron microscopy (TEM), and scanning TEM (STEM) images. The Cu-N/CF exhibited a fibrous morphology with a relatively smooth surface and no visible pores (Figure 2a-d, Figure S4-S7). In the Cu-N/PCF, the macrophase separation of the PAN and PMMA blend results in the formation of non-uniform and large PMMA domains over the PAN matrix, which act as pore-forming reagents and can be easily removed through oxidization at low temperatures in air. As a result, the Cu-N/PCF displayed discontinuous and non-uniform macro/mesopores (Figure 2e-h). In contrast, the microphase separation of PAN-*b*-PMMA directs the generation of continuous and uniform PMMA domains in the PAN matrix,³⁴ thereby creating the IPCF for supporting isolated Cu atoms with dense and interconnected mesopores throughout the surface and cross-section (Figure 2i-l).

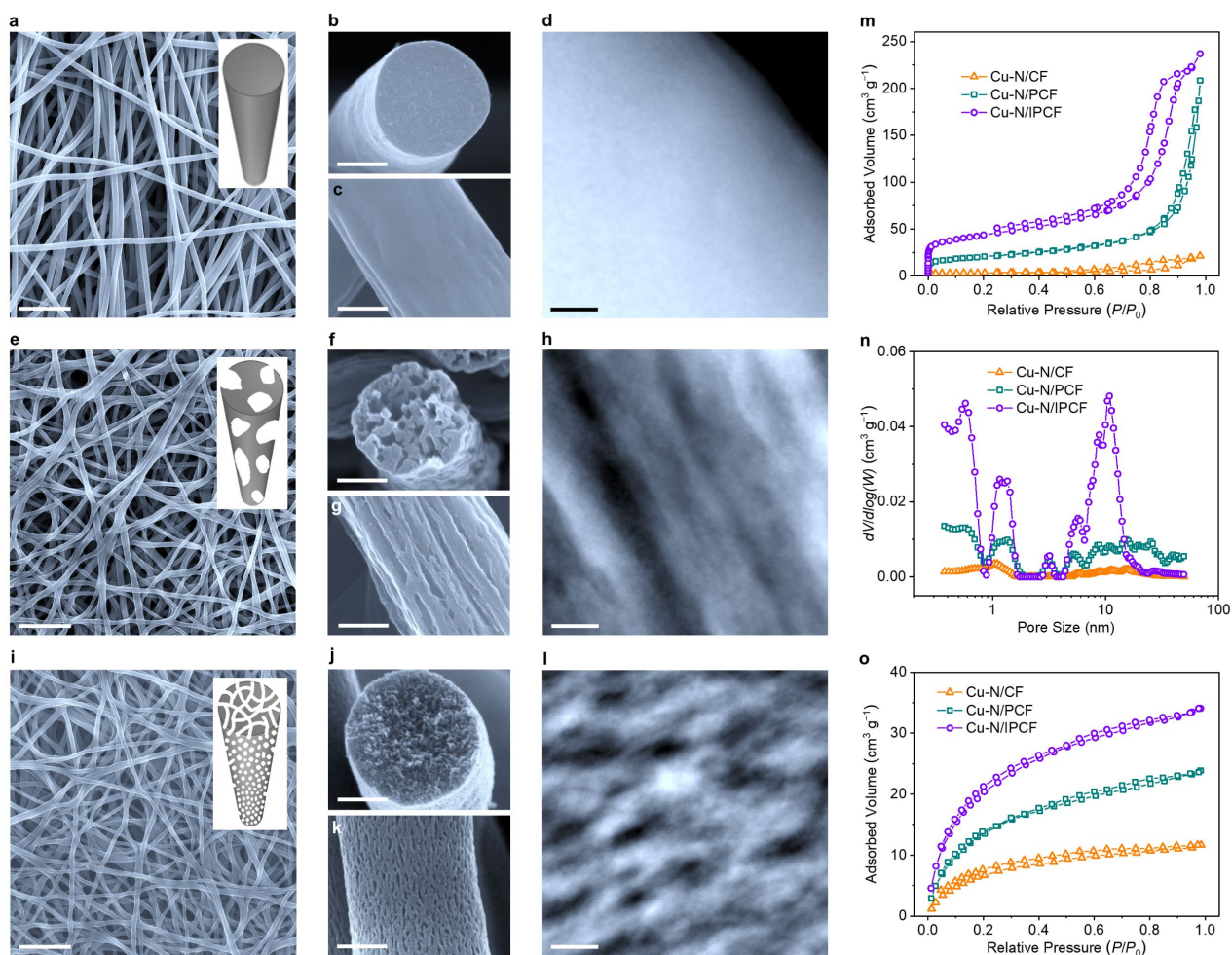


Figure 2. Characterization of morphology and pore structure. SEM and STEM images of Cu-N/CF (a-d), Cu-N/PCF (e-h), and Cu-N/IPCF (i-l). (m) N₂ gas adsorption-desorption isotherms, (n) pore size distribution, and (o) CO₂ adsorption-desorption isotherms for these catalysts. Scale bar: 5 μm (a, e, i), 200 nm (b, c, f, g, j, k), 20 nm (d, h, l).

The evolution of different pore structures was further monitored by N₂ gas adsorption-desorption isotherm measurements. The Cu-N/IPCF exhibited distinguishable features with pronounced type H1 hysteresis confirming the existence of mesopores (Figure 2m), which was not observed with the Cu-N/CF and Cu-N/PCF samples. As for the pore size distribution, the Cu-N/IPCF showed a considerable number of mesopores with a narrow distribution centered at ~ 9.3 nm (Figure 2n), whereas there are insignificant mesopores with broad distribution in the Cu-N/PCF and fewer mesopores in the Cu-N/CF. Further, the Cu-N/PCF displayed the largest surface area and CO₂ adsorption amount than the other two counterparts (Figure 2o, Table S1). These results validate the success in the synthesis of Cu-N/IPCF with uniform mesopores via the block copolymer-assisted method, which is prominently

distinct from porosities of conventional Cu-N/PCP and Cu-N/CF prepared using a PAN+PMMA blend and PAN alone, respectively. The long-range mesopores could provide efficient CO₂ transport and prevent CO intermediates from quickly escaping due to the confined diffusion channels. The unique porous microenvironment favors the creation of CO₂-rich active surfaces for desirable deep reduction to CH₄.

Atomic-level structure of single Cu sites in catalysts

We further elucidated the atomic-level structure of atomically dispersed Cu sites and their interaction with the carbon support. All catalysts showed a broad feature at $\sim 25^\circ$ in X-ray diffraction (XRD) patterns (Figure S8), suggesting no Cu-containing crystalline species on the partially graphitized carbon support, in agreement with high-resolution TEM images and TEM diffraction patterns (Figure S9). Energy-dispersive spectroscopy elemental mapping images confirm the homogeneous distribution of Cu species in the N-doped carbon matrix (Figure 3a). The actual loadings of Cu in the three catalysts were determined to be around 11 wt% by inductively coupled plasma mass spectrometry (ICP-MS) (Table S1). The aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images display dense bright spots with a size of around ~ 0.22 nm corresponding to isolated Cu atoms (Figure 3b, 3c, Figure S10, S11). X-ray photoelectron spectroscopy (XPS) analyses show a high N content of around 14.5 at.% in the surface layer of catalysts (Table S1). Deconvolution of the high-resolution N 1s spectra confirms the presence of pyridinic N (398.3 eV), pyrrolic N (400.1 eV), graphitic N (401.5 eV), and oxidized N (403.1 eV) (Figure 3d). Compared to the Cu-free N-doped IPCF (N-IPCF), the position of pyridinic N shifted to higher binding energy with an increased percentage over Cu-containing catalysts (Figure S12), while the positions of other N species remained unchanged. This observation suggests that the stabilization of single Cu metals involves strong coordination with pyridinic N forming Cu–N bonds due to the existence of lone-pair electrons on pyridinic N.³⁵⁻³⁷

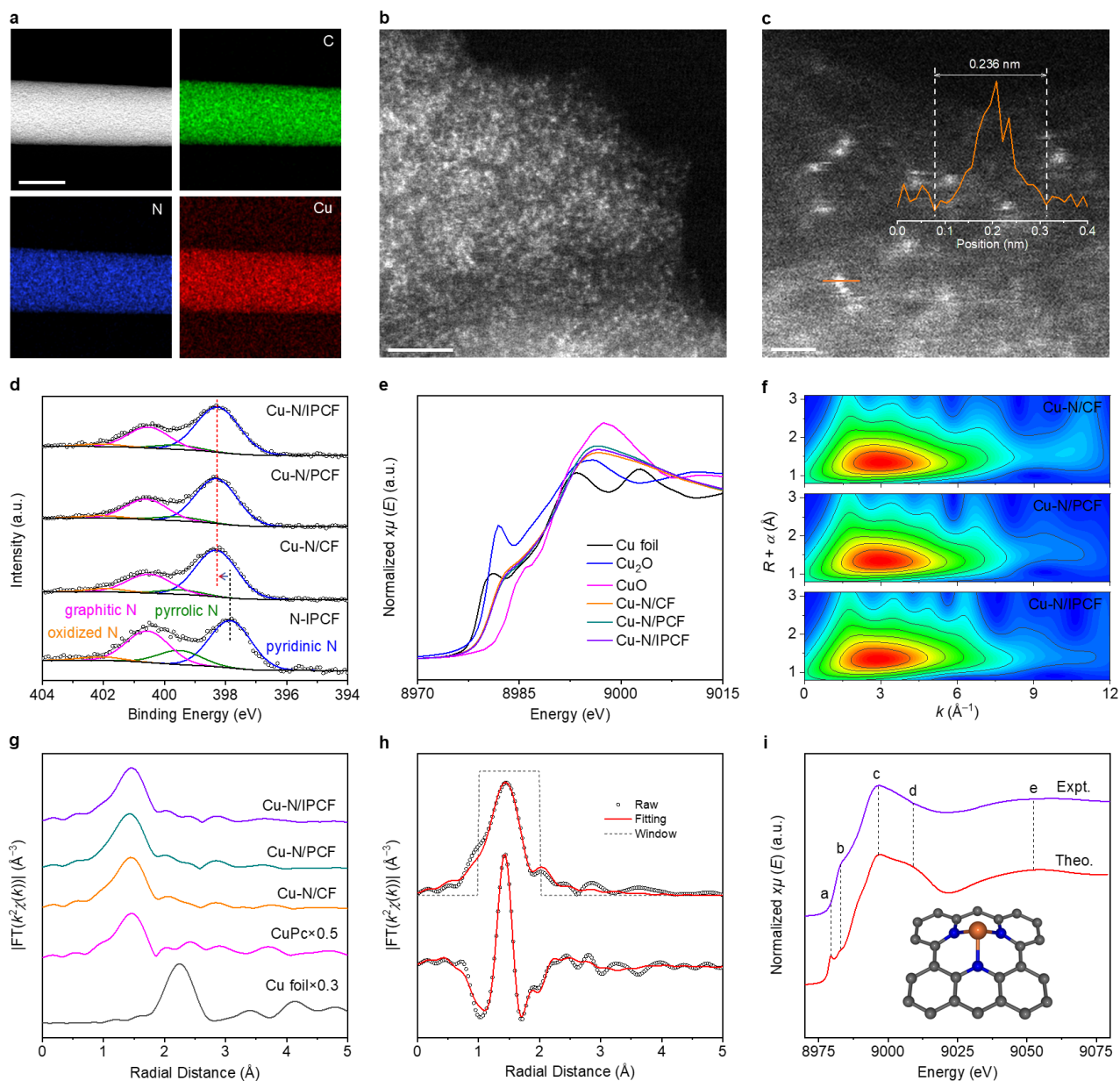


Figure 3. Atomic-level structural characterization of Cu. (a) Energy-dispersive spectroscopy elemental maps and (b-c) HAADF-STEM images for Cu-N/IPCF. (d) N1s XPS, (e) Cu K-edge XANES, (f) Wavelet transforms of EXAFS, (g) Fourier transforms of EXAFS spectra of catalysts and reference samples. (h) Fitting of EXAFS spectrum for Cu-N/IPCF. (i) Comparison between the experimental Cu XANES spectrum and the theoretical spectrum of the out-plane Cu–N₃ model. Scale bars: 400 nm (a), 2 nm (b), 0.5 nm (c).

The electronic states and local coordination of atomic Cu were further probed by using Cu K-edge X-ray absorption spectroscopy (XAS). In the X-ray absorption near edge spectroscopy (XANES), the edge positions of the three catalysts were located between the Cu₂O and CuO (Figure 3e, Figure S13). The Cu oxidation state was estimated to be +1.25 based on the correlation between the Cu oxidation state and the absorption edge positions. The Cu 2p and LMM Auger spectra further

confirm the presence of Cu^+ species (Figure S14). The Fourier transformed (FT) extended X-ray absorption fine structure (EXAFS) presents a notable peak at $\sim 1.45 \text{ \AA}$ (Figure 3g), in accordance with the first Cu–N shell of the CuPc standard; no observable Cu–Cu bond at $\sim 2.35 \text{ \AA}$ (Cu foil) was detected. The Wavelet transformed (WT) EXAFS analyses show only one maximum intensity at approximately $3.0 \text{ \AA}^{-1}$ associated with the single-metal Cu–N pair in the absence of crystalline species (Figure 3f).³⁸⁻³⁹ The fitting of EXAFS spectra unveils that the coordination number (CN) of Cu–N is around $3.18 \sim 3.31$ for these catalysts (Figure 3h, Figure S15, Table S2), indicating that the isolated Cu is coordinated with approximately three pyridinic N atoms in the first shell forming Cu–N₃ moieties. The result is consistent with the XANES and reported finding that the oxidation state of Cu species in Cu–N₃ is Cu^+ .⁴⁰ The precise atomic-level configuration of Cu–N₃ was studied by comparing the experimental Cu K-edge XANES spectra with theoretical spectra of models simulated based on the FDMNES program, where two commonly proposed models were considered, out-plane Cu–N₃ and in-plane Cu–N₃.^{20, 40} The simulated XANES result of the out-plane Cu–N₃ could fit the main features of experimental XANES spectrum much better than that of the in-plane Cu–N₃ (Figure 3i, Figure S16), suggesting that the out-plane Cu–N₃ is more likely the dominant structure in the as-prepared catalysts. Together, the combination of microscopic HAADF-STEM and spectroscopic XAS analyses unambiguously validates the well-defined, atomically dispersed, and N-coordinated Cu–N₃ sites at a high mass loading supported on three different types of carbon supports.

Experimental evaluation of CO₂ reduction

In CO₂ electrolysis experiments, catalysts were deposited onto a hydrophobic GDL to increase the local concentration of gaseous CO₂ for achieving industry-level current density (Figure S17). CO₂ reduction performance was measured in a flow cell with an alkaline electrolyte under constant-current testing conditions. We detected four gaseous products (H₂, CO, CH₄, and C₂H₄) and four liquid-phase products (HCOO[−], CH₃OH, C₂H₅OH, and CH₃COO[−]) from the direct CO₂ reduction based on the ¹³CO₂ isotope labeling experiment (Figure S18-20). Among them, HCOO[−] exhibits significantly

lower Faradaic efficiency (FE, < 2%) than others. This can be attributed to the enhanced reactivity of the Cu site to form C-bonded *COOH rather than O-bonded *OOCH intermediate; the former drives CO production and deep CO reduction to CH₄ and C₂ products through hydrogenation and C–C coupling, whereas the later produces HCOO[−] as the final product.^{41–43} Figure 4a shows FEs at current densities from 50 to 500 mA cm^{−2}. The Cu-N/CF achieved a maximum overall CO₂RR FE of 40.6% at 100 mA cm^{−2}, which decreases drastically at larger current densities with significantly increased FEs for the HER. On the contrary, the HER was successfully suppressed on porous catalysts with remarkably increased CO₂RR: HER FE ratios (Figure S21), achieving the highest overall CO₂RR FE of 55.2% at 200 mA cm^{−2} on the Cu-N/PCF and 87.8% at 300 mA cm^{−2} on the Cu-N/IPCF.

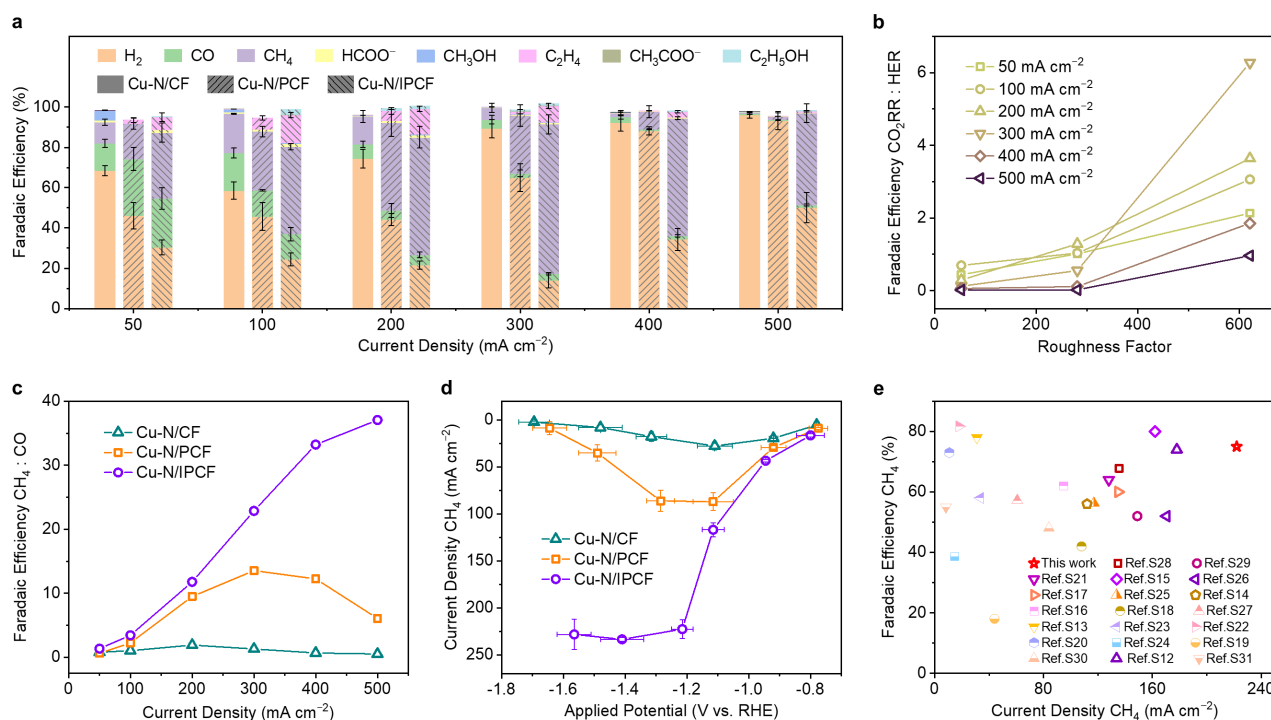


Figure 4. Electrochemical CO₂ reduction performance. (a) FEs of various products, (b) Correlation between RF and FE ratios of CO₂RR: HER. (c) FE ratios of CH₄: CO at various current densities for Cu-N/CF, Cu-N/PCF, and Cu-N/IPCF. (d) Partial current densities of CH₄ for Cu-N/CF, Cu-N/PCF, and Cu-N/IPCF. (e) Activity comparison between Cu-N/IPCF and typical Cu-based catalysts reported in the literature.

The inhibition of the HER can be attributed to the highly porous structure. In a previous study, Surendranath et al.⁴⁴ demonstrated the mesostructure-induced transport limitations in the HER suppression on porous electrodes, which can be quantified using the roughness factor (RF). We

calculated the RFs of our catalysts by normalizing the electrochemically active surface area of carbon support to the geometric area of the electrode (Figure S22, S23), showing 53.3, 280.5, and 621.2 for the Cu-N/CF, Cu-N/PCF, and Cu-N/IPCF, respectively. We noted that the increase in RF leads to an enhancement in FE ratios of CO₂RR : HER (Figure 4b), along with tendencies of increase in CO₂RR current density and decrease in HER current density (Figure S24), suggesting a strong dependence on CO₂ reduction on the porous structure of the electrode. Specifically, compared to the H₂O-rich surface on the less-porous Cu-N/CF, the porous Cu-N/PCF catalyst provides diffusion channels to deliver CO₂ to the catalyst surface within pores, while H₂O penetration is inhibited due to the hydrophobic nature of the GDE (Figure 1c). Such mass transport gradient can be substantially enhanced on the Cu-N/IPCF because of its denser long-range mesopores (Figure 1d), thereby building a CO₂-rich yet H₂O-depleted environment and enabling efficient CO₂ reduction at large currents.

We then analyzed the catalytic selectivity and activity of CO₂ reduction. CO and CH₄ are the main products of these Cu-containing catalysts. In contrast, there are only trace amounts of CO₂RR-derived products on Cu-free N-IPCF (Figure S25), verifying the highly active nature of Cu components toward CO₂ reduction. Calculating the FE ratio of CH₄ : CO suggests that block copolymer-derived mesopores are essential for promoting deep CO transformation to CH₄, with the trend following the order of Cu-N/IPCF > Cu-N/PCF > Cu-N/CF, particularly at high current densities (Figure 4c). This observation can be ascribed to the steric nanoconfinement effect that enrich *CO intermediate within bicontinuous mesopores.^{30-31, 45} This can be corroborated by *in-situ* Raman spectroscopy (Figure S26), in which the bands at around 286 and 382 cm⁻¹ correspond to the frustrated rotation of adsorbed *CO and Cu-*CO stretching, respectively.⁴⁶⁻⁴⁷ The surface-absorbed *CO appeared on the Cu-N/IPCF upon applying potentials negative than -1.0 V, which, however, was not observed on both Cu-N/PCF and Cu-N/CF. Therefore, compared to the Cu-N/CF and Cu-N/PCF, the Cu-N/IPCF could afford the most enhanced nanoconfinement contribution to prevent CO intermediates from escaping long-range mesochannels, considerably improving its residence time and successive reduction to CH₄ on the Cu sites before becoming gaseous CO (Figure 1b-d). Note that

although such local CO-rich characteristics also contribute to the improvement in FE ratios of C₂:C₁ on the Cu-N/IPCF (Figure S27), the intrinsically better CO₂-to-CH₄ reactivity than C–C coupling on the *in-situ* reconstructed Cu site guarantees the enhanced selectivity of CH₄ than C₂ products, as supported by the following *in-situ* XAS and DFT studies.

A maximum CH₄ FE of 74.2% was achieved on the Cu-N/IPCF at –1.21 V, much higher than 43.4% on the Cu-N/PCF and 19.5% on the Cu-N/CF at –1.11 V. In addition, the Cu-N/IPCF exhibited significantly larger partial current density than the Cu-N/CF and Cu-N/PCF at applied potentials negative than –1.11 V (Figure 4d), due to the inhibition in the hydrogen evolution rate. As a result, high-efficient CH₄ generation yields an industry-level partial current density of 222.6 mA cm^{–2} at –1.21 V. Remarkably, the performance of the Cu-N/IPCF places it at the top level among reported Cu-based catalysts (Figure 4e, Table S3), despite there still needs significant performance improvement to meet the application threshold for profitable electricity-powered CO₂-to-CH₄ conversion.⁴⁸ The catalyst can maintain durable CO₂-to-CH₄ electrolysis at 300 mA cm^{–2} for three hours (Figure S28a), during which the slight decrease in CH₄ FE after two hours is likely as a consequence of the declined surface hydrophobicity of GDE (Figure S28b).

Identification of atomic-level Cu site structure under CO₂RR

To understand the actual catalytically active sites, we performed *in-situ* XAS to investigate the possible structure evolution of single Cu metal sites in the optimal Cu-N/IPCF catalyst under operating CO₂ reduction conditions. The oxidization state of Cu remained unchanged under open circuit voltage (OCV) and dry conditions based on XANES spectra (Figure 5a) and first derivative profiles (Figure S29). However, it decreased from +1.25 to +1.06 when a negative potential of –0.8 V was applied. Further sweeping potential from –1.2 to –1.6 V resulted in the continuous reduction to +0.89. These results reveal the unstable characteristics of atomically dispersed Cu sites and the occurrence of electrochemical Cu^{δ+} reduction accompanied by CO₂RR.

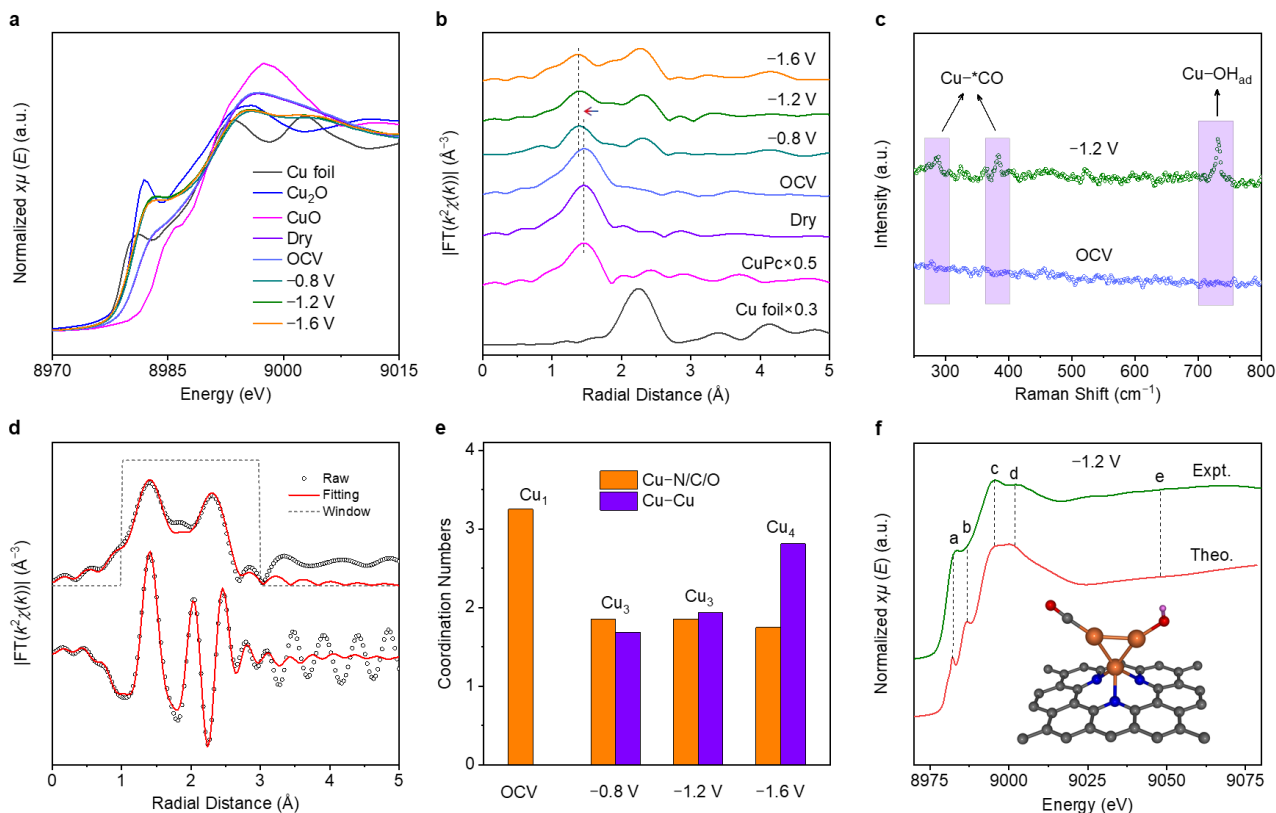


Figure 5. *in-situ* characterization of active site structure. (a) *in-situ* Cu K-edge XANES, (b) Fourier transforms of EXAFS, and (c) *in-situ* Raman spectra of the Cu-N/PCF. (d) Fitting of EXAFS spectrum obtained at -1.2 V. (e) Fitted CNs for Cu-N/C/O and Cu-Cu coordinations at different potentials. (f) Comparison between the experimental Cu XANES spectrum at -1.2 V and the theoretical spectrum of the Cu₃-N₃-CO₁-OH₁ model. The brown, gray, blue, red, and pink balls represent Cu, C, N, O, and H atoms, respectively.

We continued exploring Cu structures by investigating EXAFS spectra under dynamic electrochemical potentials. Similar to the fresh catalyst, no Cu-Cu coordination was observed under the OCV (Figure 5b). At -0.8 V, the Cu-Cu bond appeared at 2.28 Å, accompanied by a decrease in the intensity of Cu-N relative to that at the OCV. Such transformations were intensified from -1.2 to -1.6 V, confirming the collapse of Cu-N₃ moieties and their dynamic reconstruction to metallic Cu species during the CO₂ reduction. Notably, the radial distance of Cu-N coordination under the realistic CO₂ reduction conditions declined by 0.07 Å compared to that at the OCV. The observation suggests the change in the first-shell coordination atoms towards Cu. In general, the first-shell coordinated atoms could be anchoring atoms from the carbon support, adsorbed species from the KOH electrolyte, and intermediates from CO₂RR. In the N-doped carbon, N dopants possessing

electron-withdrawing properties have stronger interactions with Cu than C atoms.^{16, 25} Thus, N dopants on the IPCF support might be the preferable anchoring sites to immobilize metallic Cu species dynamically formed from isolated Cu atoms,^{25, 49} as manifested by the calculated formation energy (Figure S30). The presence of adsorbed *CO intermediate and OH_{ad} groups can be confirmed by *in-situ* Raman spectrum (Figure 5c),⁵⁰⁻⁵¹ in which the *CO is from CO₂ reduction and the OH_{ad} can be ascribed to the specific adsorption of weakly solvated OH⁻ ions from the electrolyte. Thereby, the declined average bond length of Cu–N might be a result of the addition of Cu–*CO and Cu–OH_{ad} bonding, generating Cu–N/C/O coordination over the metallic Cu species.

EXAFS spectra were further fitted to resolve the CNs. As the first-shell coordinated N/C/O atoms in the fitting cannot be distinguished, Cu–N and Cu–Cu paths were included in the EXAFS fitting using Cu as the core (Figure 5d, Figure S31, Table S4). The CNs of Cu–N is 3.25 at the OCV (Figure 5e), implying that Cu–N₃ remains stable in the CO₂-saturated electrolyte without CO₂RR. Notably, the CNs changed drastically upon applying CO₂ reduction potentials, showing 1.87/1.68, 1.85/1.94, and 1.75/2.81 for Cu–N/C/O and Cu–Cu at –0.8, –1.2, and –1.6 V, respectively. The results suggest that single-atom Cu metals were transformed into the Cu₃ cluster at –0.8 and –1.2 V, which turned into the Cu₄ cluster with the potential down to –1.6 V. We further performed XANES simulations to reveal the atomic-level structure of Cu₃ at –1.2 V where the best CO₂-to-CH₄ performance was achieved. Given the configuration of N dopants, fitted CN for Cu–N/C/O, and *in-situ* Raman results, three possible structures were considered, including Cu₃-N₃-CO₁-OH₁, Cu₃-N₃-CO₂-OH₁, and Cu₃-N₃-CO₁-OH₂ (Figure 5f, Figure S32). Among them, the calculated XANES spectrum of the Cu₃-N₃-CO₁-OH₁ shows the best match with the experimental spectrum at –1.2 V than others, suggesting that the Cu₃-N₃-CO₁-OH₁ might be the dominant configuration derived from the *in-situ* reconstruction of Cu–N₃ under CO₂ reduction at –1.2 V. Note that the theoretical CN of Cu–N/C/O over the Cu₃-N₃-OH₁-CO₁ is 1.67, close to the experimental value of 1.85. The structure also coincides with the fact that the oxidation state of electrochemically reduced Cu species is not zero due to strong electron transfer from the metallic Cu₃ cluster to surrounding N-doped carbon, *CO intermediates, and

adsorbed OH_{ad} species, leaving cationic Cu moieties. We have also characterized the spent catalyst by *ex-situ* methods, displaying the single atomic characteristics of Cu species (Figure S33, S34). The observation suggests the reversion of *in-situ* formed Cu₃ clusters back to isolated Cu in the absence of an electric field and/or being exposed to air, in agreement with previous findings.^{23-24, 52} In sum, it is the N₂OH-Cu₃ cluster, instead of atomically dispersed Cu-N₃, that serve as the catalytic site for CO₂-to-CH₄ reduction at -1.2 V.

Theoretical understanding of catalytic mechanisms

Density functional theory (DFT) calculations were performed to provide mechanistic insights. In the CO₂-to-CH₄ transformation, the hydrogenation of *CO to *CHO is a key step,^{4, 53-55} which requires a relatively strong *CO adsorption to inhibit its desorption becoming gaseous CO. We first considered CO adsorption affinity on the Cu atom of the N₂OH-Cu₃ and C/N atoms of the N₃-C (Figure 6a, Figure S35). The Cu site shows a negative adsorption energy of -0.76 eV, indicative of favorable CO adsorption. Yet, positive adsorption energy suggests weak *CO binding on the N₃-C. This prediction demonstrates that the Cu atom of the N₂OH-Cu₃ is likely the catalytically active site for reducing CO₂ to CH₄ because its preferential *CO adsorption ensures deep *CO transformation. Charge density analysis displays strong electronic interaction between the *CO and N₂OH-Cu₃ with electrons transferred from the Cu atom to *CO (insert of Figure 6a), enabling stable *CO binding and subsequent PCET-driven reduction to CH₄ on the N₂OH-Cu₃ site. We further carried out density of states (DOS) analysis to unveil the bonding interaction upon the adsorption of *CO on the N₂OH-Cu₃. It is the Cu 3d_{xz}/d_{x2} orbitals that participate in orbital coupling with C 2p of *CO adsorbate based on their orbital overlapping (Figure 6b, Figure S36).

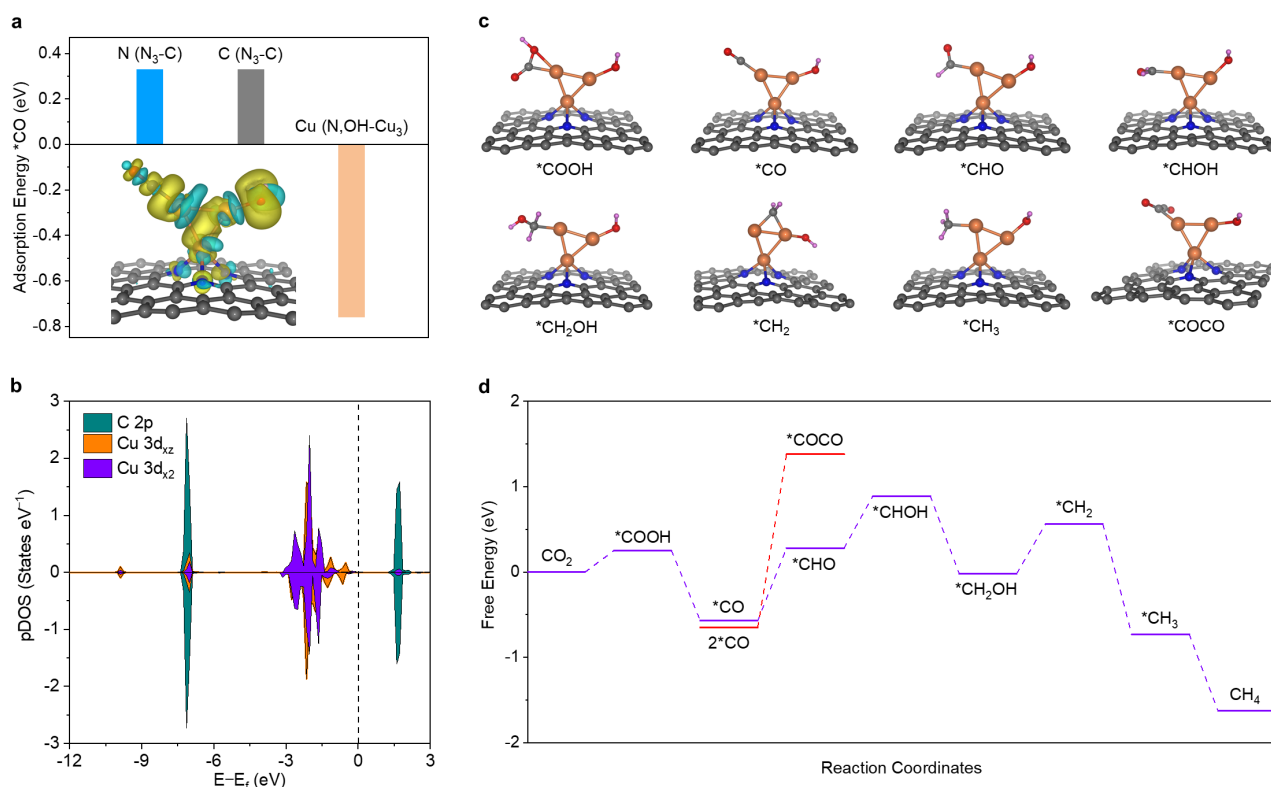


Figure 6. DFT calculations. (a) Predicated adsorption energy of *CO adsorbed on various sites. Insert is the charge density difference analysis of *CO on the N,OH-Cu_3 . Yellow zone is charge increasing and cyan zone is charge decreasing. (b) Calculated density of states showing the orbital interaction between C 2p of *CO and Cu 3d_{xz}/d_{z2} of the N,OH-Cu_3 . (c) Optimized adsorption configurations of various intermediates on the N,OH-Cu_3 . (d) Free energy diagram of CO_2 -to- CH_4 reduction and *COCO coupling on the N,OH-Cu_3 at $U = 0$ V vs. RHE.

We further calculated the free energy diagram, in which the reaction pathways involving *COOH , *CO , *CHO , *CHOH , $\text{*CH}_2\text{OH}$, *CH_2 , and *CH_3 intermediates, were considered.^{10, 54, 56} The adsorption configurations of C-bound intermediates are depicted in Figure 6c. The results show that free energy barriers are 0.25, 0.85, 0.61, and 0.58 eV for CO_2 activation to *COOH , *CO hydrogenation to *CHO , transformation of *CHO to *CHOH , and reduction of $\text{*CH}_2\text{OH}$ to *CH_2 on the N,OH-Cu_3 (Figure 6d), respectively. The other elementary steps are thermodynamically downhill. Thus, the hydrogenation of *CO to *CHO is the potential-limiting step in reducing CO_2 to CH_4 . In CO_2 reduction, forming C_2 products via C–C coupling is a grand obstacle challenging selectivity toward CH_4 . Here, the free energy barrier for *CO dimerization, a key step to form C–C bond,⁵⁷⁻⁵⁹ was calculated to be 2.03 eV, significantly larger than that of the CO_2 -to- CH_4 reduction. The results manifest the intrinsically enhanced capability in steering improved CH_4 selectivity with the

suppressed C₂ pathway on the N₂OH-Cu₃ site. The promoting role of surface-bound OH_{ad} species in CH₄ generation was also verified based on the increased free energy barrier for *CO hydrogenation on the OH_{ad}-free N-Cu₃ compared to that on the N₂OH-Cu₃ (Figure S37). Overall, the theoretical predictions support the experimental enhanced CO₂-to-CH₄ activity of the *in-situ* formed N₂OH-Cu₃ clusters derived from monodispersed Cu–N₃ moieties under the realistic CO₂RR conditions.

Conclusions

In this study, we demonstrated a catalyst design principle of integrating atomic-level tuning of Cu site's structure with mesoscale engineering of carbon support's pore architecture to steer the CO₂ reduction pathway on carbon-supported Cu catalysts. The combined experimental and theoretical investigations validated the effectiveness of this strategy toward CO₂-to-CH₄ transformation.

Especially utilizing the microphase separation of a PAN-*b*-PMMA block copolymer, a long-range interconnected mesoporous N-doped carbon fiber support was developed to host atomically dispersed and N-coordinated Cu moieties. The bicontinuous mesochannels of the carbon support create a favorable local reaction microenvironment that not only offers H₂O-deficient and CO₂-rich interface in the pore region suppressing hydrogen evolution but also builds an enhanced nanoconfinement space with an extended transport path for CO intermediate facilitating deep CO reduction. More importantly, *in-situ* electrochemical XAS and Raman spectroscopy demonstrated a unique dynamic reconstruction of atomically dispersed Cu sites to Cu₃ clusters immobilized by nitrogen of the carbon support and hydroxyl species from the electrolyte under the operating CO₂ reduction conditions, serving as the actual catalytic site for efficient CO₂-to-CH₄ reduction. DFT calculations further predicted that the N₂OH-Cu₃ clusters selectively drive CH₄ formation with the suppressed C₂ pathway along an energetically favored CO₂ → *CO → *CHO → *CH₂OH → CH₄ route. Under the synergy of favorable *in-situ* reconstructed N-Cu₃ sites and long-range confined mesoreactors, we achieved a CH₄ Faradaic efficiency of 74.2% at an industry-level current density of 300 mA cm⁻², showing great promises for practical CO₂-to-fuel electrolysis implementation. The

rational design concept could pave an effective platform for developing a broad of advanced catalysts with optimal structures at multiscale, capable of regulating reaction pathways and selectivity for complex catalytic systems that have multiple steps and reaction routes.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at

Detailed experimental procedures, characterization methods, computational methods, extended figures and tables (PDF)

Conflict of interest

The authors declare no competing financial interest.

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